

Video 1: Statistical Power, Clearly Explained!!!

Video Link: <https://youtu.be/Rsc5znwR5FA>

Channel: StatQuest with Josh Starmer

I. Introduction & Prerequisites

- **Topic:** This video provides a clear explanation of Statistical Power.
- **Assumed Knowledge:** To understand this, you should already be familiar with:
 - i. **P-values**
 - ii. **The Normal Distribution** (a "bell curve")

II. High Power Example: When the Difference is Obvious

- **Scenario:** Imagine comparing the weights of two groups of mice:
 - i. **Group 1 (Left):** Mice on a special diet (they weigh less).
 - ii. **Group 2 (Right):** Mice on normal mouse food (they weigh more).
- **Observation:** The two distributions (bell curves) are very different and have **very little overlap**. It's easy to see the difference between the diets.
- **Experiment:** We take a small set of measurements (a small sample) from each group.
- **Result:** When we compare the means (averages) of these two samples, we get a **very small p-value**, for example, **$p = 0.0004$** .
- **Conclusion:**
 - A small p-value (typically < 0.05) tells us to **reject the null hypothesis**.
 - **The Null Hypothesis:** This is the default idea that there is *no* difference—that both sets of data actually came from the same, single distribution (e.g., a "middle" distribution).
 - Our small p-value allows us to **correctly reject** this null hypothesis and conclude that the diets *do* have a different effect.
- **Defining High Power:**
 - If we repeated this experiment many times, we would have a **high probability** of getting a small p-value every time.
 - This "high probability of correctly rejecting the null hypothesis" **is the definition of high power**.
 - Alternatively, power is the probability that we will correctly get a small p-value.
- **When Power Fails (Even in a High-Power_Scenario):**
 - Every once in a while, just by random chance, we might get a "fluke" sample where the mice from the special diet group happen to be a bit heavier, and the normal diet mice happen to be a bit lighter.

- This would lead to a **large p-value** (> 0.05). In this case, we would **fail to reject** the null hypothesis, even though we know a real difference exists. This is an error, but it's not likely to happen when power is high.

III. When Does Power Not Apply?

- If there was **zero difference** between the two diets (i.e., both groups of mice truly came from the *exact same* distribution), the null hypothesis would be **true**.
- Since the null hypothesis is true, it's impossible to "correctly reject" it. Therefore, the concept of "power" (the probability of *correctly* rejecting) doesn't apply in this situation. Power only applies when there is a *real* difference to find.

IV. Low Power Example: When the Difference is Small

- **New Scenario:** Imagine a special diet that *does* make a difference, but it's "not very good".
- **Observation:** The two distributions are still distinct, but they have a **huge amount of overlap**. It's very hard to see the difference.
- **Experiment:** We take a **small sample size** (e.g., 3 mice from each group).
- **Result:** When we compare the means, we get a **large p-value**, for example, **$p = 0.34$** .
- **Conclusion:**
 - Because $p > 0.05$, we **fail to reject the null hypothesis**.
 - This is a **bummer**, because we *know* a real difference exists, but our experiment failed to find it.
- **Defining Low Power:**
 - If we repeated this experiment many times (with a small sample size), *most* of the time we would get a large p-value and fail to find the real effect.
 - Only occasionally, by sheer luck, would our samples be different enough to give a small p-value.
 - When there is a lot of overlap AND we use a small sample size, we have a low probability of correctly rejecting the null hypothesis. This is **low power**.

V. How to Fix Low Power

- **The Solution:** The good news is that we can **always increase power by increasing the sample size**.
- **Power Analysis:** A **power analysis** is a statistical tool that tells you *how many* measurements (what sample size) you need to collect to have a good amount of power for your experiment.

VI. Final Summary

1. **Power** = The probability that we will **correctly reject the null hypothesis** (i.e., the probability of finding a real effect when one exists).
2. **Little Overlap** (big difference) = **High Power**.
3. **Lots of Overlap** (small difference) + **Small Sample Size** = **Low Power**.
4. **To Increase Power** = **Increase the Sample Size**.
5. **Power Analysis** = A tool that tells you what sample size you need.

Video 2: Power Analysis, Clearly Explained!!!

Video Link: https://youtu.be/VX_M3tlyiYk

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I. Introduction & The "P-Hacking" Problem

- **Topic:** This video explains what a Power Analysis is and how to do one.
- **Assumed Knowledge:** You should be familiar with Power, p-hacking, and the difference between population parameters and estimated parameters.
- **The Problem (A "Tempting" Scenario):**
 - **Experiment:** We test two drugs (A and B) to treat a virus, using 3 people for each drug.
 - **Observation:** The data *looks* like Drug A is better (people recover faster).
 - **Result:** We run a statistical test, and the **p-value = 0.06**.
 - **Dilemma:** Our significance threshold is 0.05. Since 0.06 is > 0.05 , we **cannot reject the null hypothesis**. We must conclude that we *did not* find a statistically significant difference.
 - **The Temptation:** It's *so tempting* to think, "Oh, 0.06 is so *close*! Let's just add one more person to each group and re-run the test".
 - **This is P-HACKING.** It's a "statistical sin." You cannot repeatedly add data until you get the p-value you want.

II. The Solution: What a Power Analysis Does

- **The Right Thing to Do:** Instead of p-hacking, you should stop, and *before you run the experiment again*, you should do a **power analysis**.
- **Definition:** A power analysis is a calculation that determines what **sample size** you will need for your *next* experiment to ensure you have a **high probability** (i.e., high power) of correctly rejecting the null hypothesis.

- **The Benefit:** If the power analysis says you need (for example) 9 people per group, you run the *new* experiment with 9 people per group. Whatever p-value you get at the end, you can be confident that you used *enough data* to make a good decision.

III. Why Sample Size is the Key to Power

- **Main Factors in Power:**
 - i. **The Overlap** between the distributions (the size of the effect you're looking for).
 - ii. **The Sample Size** (how much data you collect).
- The more overlap there is (the smaller the effect), the larger the sample size you will need to achieve the same amount of power.
- **Why? It's About Averages (Means):**
 - In an experiment, we don't compare all the individuals. We compare **summaries**, like the **mean** (average) of each group.
- **Example 1: Sample Size = 1**
 - If you take just *one* measurement and use it to estimate the "population mean" (the true average), your estimate is just that one measurement.
 - If you get a "wonky" point (an outlier), your *estimated mean* is also "wonky".
 - If you do this repeatedly, your estimated means will be "all over the place". There is **high variation** and **low confidence** in your estimates.
 - When you compare two groups with $n=1$, the estimated means are so unreliable that they will overlap a lot, leading to a large p-value and **low power**.
- **Example 2: Sample Size = 2**
 - Now, you take *two* measurements at a time and use their *average* as the estimated mean.
 - If you get one "wonky" point, the other point will likely be more normal and will "compensate," pulling the average back toward the center.
 - Extreme measurements have *less effect*.
 - The resulting estimated means are closer to the true population mean. There is **less variation** and **more confidence**.
- **Example 3: Sample Size = 10**
 - If you take 10 measurements and average them, the effect of any single "wonky" point is "drowned out".
 - The estimated means will be *very close* to the *true* population mean.
 - **The Big Idea:** Even if the *original distributions* for Drug A and Drug B overlap a lot, the *distributions of the estimated means* (from $n=10$ samples) will be very narrow and will **not overlap**.
 - Because the *means* are now clearly different, you will have a **high probability** of getting a small p-value and correctly rejecting the null hypothesis. **This is high power**.

- **Note:** This principle works for *any* shape of distribution (not just normal curves) because of the **Central Limit Theorem**.

IV. How to *Actually* Do a Power Analysis (The 3 Inputs)

A power analysis tells you the **sample size** you need. To calculate it, you must first provide three pieces of information:

1. **Power (The Goal):** How much power do you want?

- This is your desired probability of finding a real effect.
- A common, arbitrary value is **0.8** (an 80% chance to correctly reject the null).

2. **Significance Threshold (Alpha):** What is your p-value cutoff?

- This is the risk you're willing to take of "finding" a difference that isn't real (a Type I error).
- A very common value is **0.05**.

3. **Effect Size (The Overlap):** How big of a difference are you looking for?

- This is the hardest part. It's a single number that combines the *distance between the means* and the *standard deviations* (the spread) of the two groups.
- **How to estimate it?** You must make an educated guess *before* the experiment, using:
 - **Prior Data:** (Like our $p=0.06$ experiment).
 - **Literature Search:** See what effect sizes other researchers have found for similar experiments.
 - **An Educated Guess:** What is the *smallest difference* that would be clinically or practically meaningful to you?
- **Example Calculation:**
 - From our prior data: The difference in means was 10. The standard deviations were 7 and 6.
 - We can "pool" the standard deviations to get an average of 6.5.
 - Effect Size (d) = (Difference in means) / (Pooled standard deviation) = $10 / 6.5 = 1.5$.

V. The Final Step: The Calculator

1. You now have your three inputs:

- **Power = 0.8**
- **Alpha = 0.05**
- **Effect Size = 1.5**

2. Google "statistics power calculator" (many universities host free ones).

3. Plug in your three numbers.

4. The calculator will output the required **Sample Size**.

5. **Result:** In this example, the calculator says **"sample size = 9"**.

6. **This means you need 9 people *per group*** (9 for Drug A, 9 for Drug B) to have an 80% chance of finding a statistically significant difference ($p < 0.05$), *assuming* the true effect size is 1.5.